EXPERIMENT 08 — LAB RECORD

Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application

Design and Implementation of User Permissions, Roles, RoleHierarchy, and DataSharing

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Aim

To configure user management and role-based access control (RBAC) within the Library Book Issue Request Form application built in Zoho Creator, by creating permission sets, constructing a role hierarchy, and configuring a custom data-sharing rule that grants the library admin role Read/Write access to book issue request data owned by the student role.

Objectives

- To review the application's live dashboard, which summarises book issue requests by book category, number of copies and member type.
- To create two permission sets, Library Administrator and Library Member, alongside the default Read and Write sets under User Permissions.
- To construct the role hierarchy CEO → {library admin, student, Faculty, Staff}.
- To configure a Data Sharing rule, Student_to_Library_Admin, sharing book issue request data from the student role to the library admin role with Read/Write access.
- To understand how RBAC separates data ownership (who submits a record) from data visibility (who else may view or edit it).

Software / Tools Required

- Zoho Creator (web-based low-code application development platform)
- Library Book Issue Request Form application (created on Oct 06, 2026)
- Web browser with an active internet connection
- Zoho account with access to the application in Edit mode

Introduction / Theory

User Management in Zoho Creator

User management decides who can access an application and what each person may see and do within it. Users are added to an application and associated with permission sets (profiles) and roles; together these control access to forms, reports, pages and records. In the Settings area, the User Permissions section groups these controls under three tabs: Permissions, Roles and Data Sharing.

Role-Based Access Control (RBAC)

RBAC is an access-control model in which rights are assigned to roles instead of individual users, and users acquire rights through the roles they hold. It simplifies administration, supports the principle of least privilege, improves data security, and keeps the access policy consistent and easy to audit.

Key Concepts

Concept	*Description*
Users	Individuals added to the application who sign in to work with its forms, reports and data.
Permissions	Named permission sets (profiles) defining what a user may do with components. This application lists Library Administrator, Library Member, Read and Write.
Roles	Positions in the organisational structure (CEO, library admin, student, Faculty, Staff) that determine a user's place in the hierarchy and the visibility of records.
Role Hierarchy	A tree in which each role reports to another. A higher role can generally view the records of roles beneath it; the reverse does not apply.
Data Sharing	Share rules that extend record visibility beyond the hierarchy, from one role or user to another, with a defined level of access (Read Only or Read/Write).

Application Specifications

Parameter	*Detail*
Application Name	Library Book Issue Request Form
Primary Forms	Library management form (book issue requests)
Dashboard	Chart by book category, number of copies and member type (Student, Faculty, Staff), with Approvals pending and Requests declined summary tiles
Permission Sets	Library Administrator, Library Member, Read, Write
Role Hierarchy	CEO → {library admin, student, Faculty, Staff}
Custom Sharing Rule	Student→ library admin, on the library management form, Read/Write access

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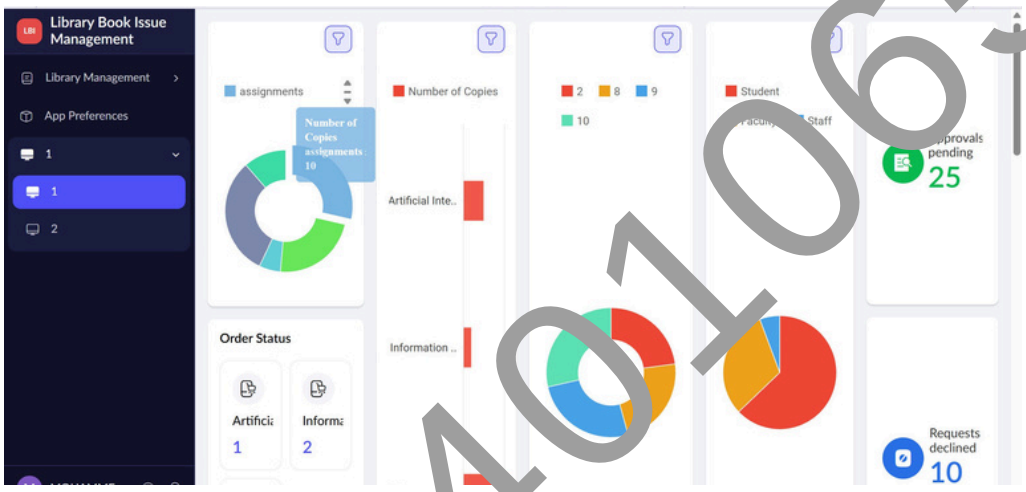
Procedure

1. Sign in and locate the application — Signing in to Zoho Creator opens the home page listing the available applications; the application card (created Oct 06, 2026), shown in Zoho Creator under the name Library Book Issue Management, is located among them.

The home page lists both applications in the account, with the Library Book Issue Management card shown first, alongside Application and Edit shortcuts.

2. Open the dashboard — Opening the application loads its dashboard in Access mode, showing charts and summary tiles for the book issue requests.

Figure 8.2: Application Dashboard (Access Mode)



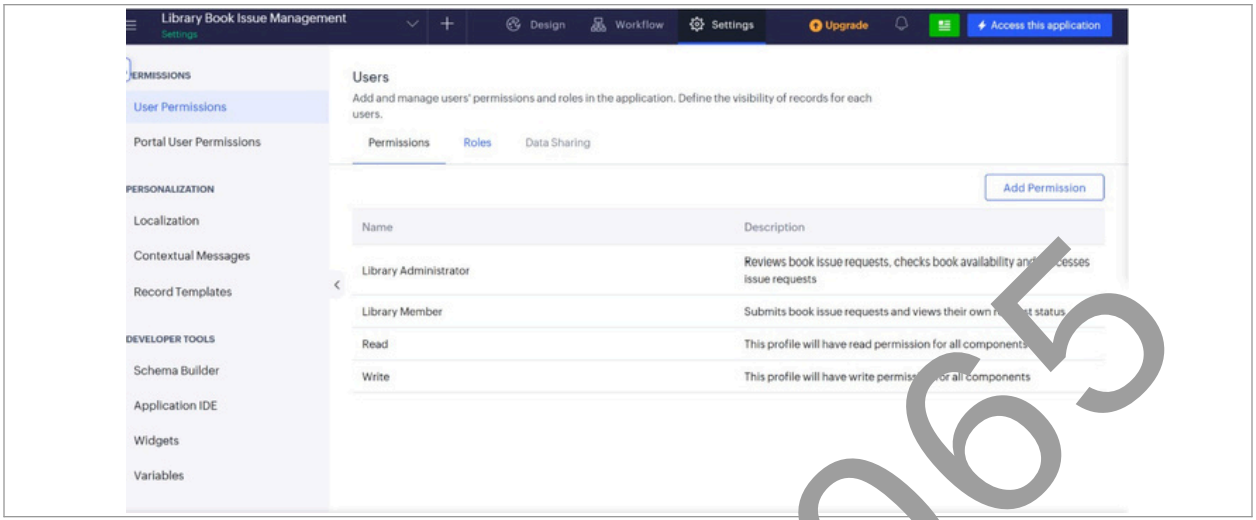
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The dashboard shows donut and pie charts and a bar chart summarising requests by book category, number of copies and member type (Student, Faculty, Staff), along with the Approvals pending (25) and Requests declined (10) summary tiles.

3. Enter Edit mode and open Settings — Selecting Edit this application switches to Edit mode, and the Settings tab is selected from the top navigation bar (Design, Workflow, Settings).

4. Open User Permissions — Under the PERMISSIONS group, User Permissions is selected, opening the Users page with the Permissions, Roles and Data Sharing tabs.

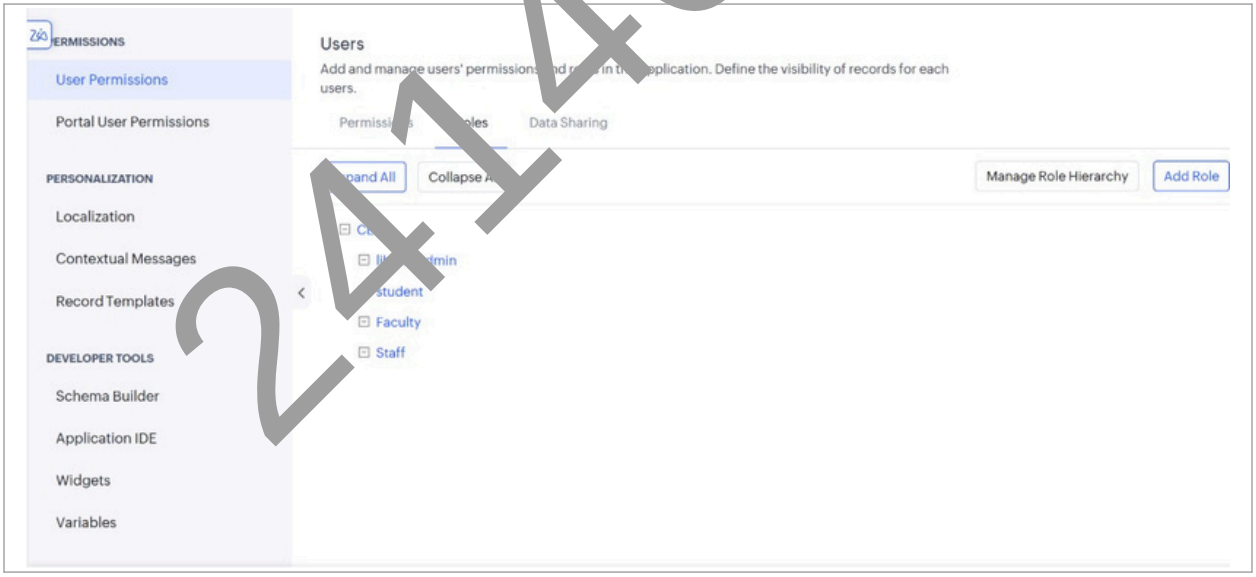
Figure 8.3: User Permissions — Permissions Section



The Permissions tab lists four permission sets: Library Administrator (reviews book issue requests, checks book availability and processes issue requests), Library Member (submits book issue requests and views their own request status), Read, and Write — each with a short description and an Add Permission option.

5. Build the role hierarchy — On the Roles tab, the role tree is arranged as CEO → library admin, student, Faculty, and Staff, using Expand All, Collapse All, Manage Role Hierarchy and Add Role as needed.

Figure 8.4: Roles and Role Hierarchy

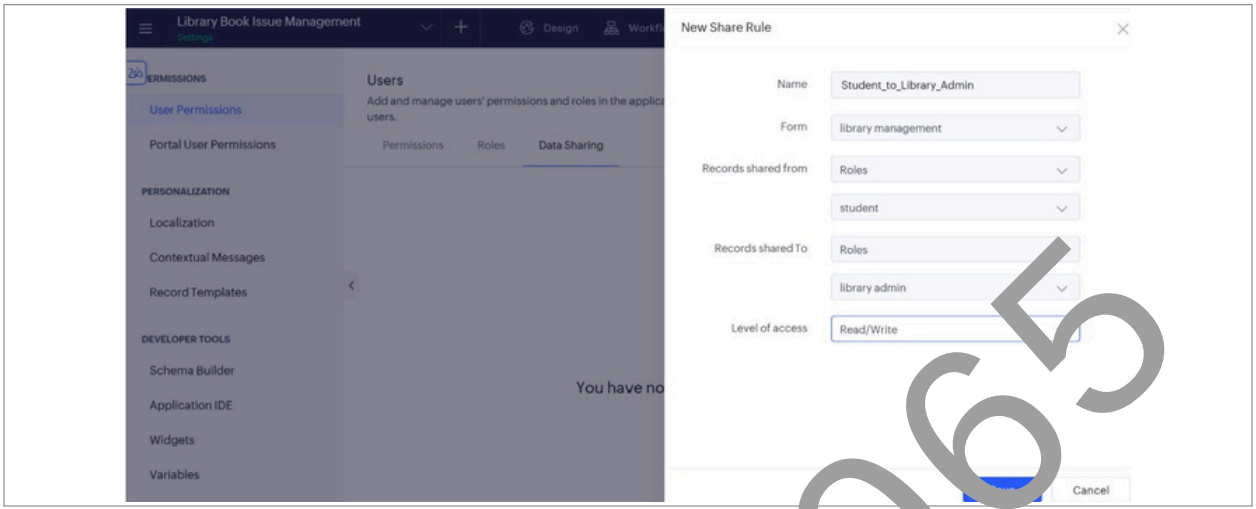


The Roles tab shows the completed hierarchy: CEO at the top, with library admin, student, Faculty, and Staff all reporting to the CEO as sibling roles.

6. Create a new share rule — On the Data Sharing tab, Add Rule is selected, the rule is named Student_to_Library_Admin, the Form is set to library management, and Records shared from is set to Roles → student.

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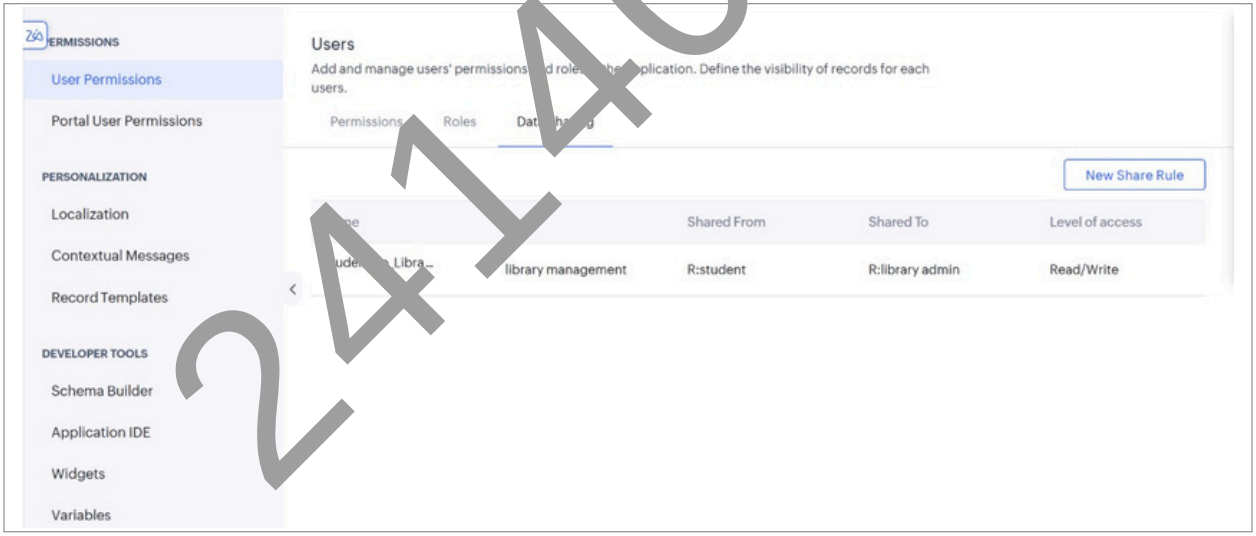
Figure 8.5: New Share Rule — Rule Configuration



The New Share Rule panel shows the rule Student_to_Library_Admin on the library management form, with Records shared from set to Roles → student, Records shared To set to Roles → library admin, and Level of access set to Read/Write.

7.Saveandverifytherule — Selecting Save adds the rule to the Data Sharing list, where the saved values can be checked.

Figure 8.6: Configured Data Sharing Rule (Student → Library Admin, Read/Write)



The Data Sharing list now shows the saved rule — Form: library management, Shared From: R:student, Shared To: R:library admin — with the Level of access set to Read/Write.

Data Sharing

The application follows a layered model: Form (library management / book issue requests) → Dashboard → Security Layer (Permissions, Roles, Data Sharing). The role hierarchy runs from the CEO (overall oversight) down to the library admin, student, Faculty, and Staff roles, which sit side by side beneath it. By default the hierarchy gives only upward visibility: higher roles see the records of the roles below them, but not the reverse.

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The custom data-sharing rule shares book issue request data from the student role to the library admin role at Read/Write level. Because the library admin and student roles are siblings under the CEO, the library admin cannot see student requests through the hierarchy alone. The rule gives the library admin an explicit, auditable Read/Write channel to review and update request details directly, keeps the scope directional (student → library admin), documents exactly who can modify what in the Data Sharing list, and lets the library admin act on records without needing elevated, system-wide Write permission.

RBAC Working

Role

Level		***Intended Function***
1 (Top)	CEO	Overall authority; able to view the records of every role reporting to it. Reviews book issue requests, checks book availability and processes the requests; gains Read/Write access to student-owned requests through the custom share rule.
2	library admin	
2	student / Faculty / Staff	Submit book issue requests and view the status of their own requests.

- Users in the student, Faculty, and Staff roles work with records according to their assigned permission set.
- Because all four roles report directly to the CEO, higher role visibility applies only to the CEO; the sibling roles do not automatically see each other's records.
- The Read/Write share rule grants the library admin a controlled, explicit channel to student-owned book issue requests.
- The permission sets (Library Administrator, Library Member, Read, Write) further limit the specific operations each user may carry out on each component.

Result

- The Library Book Issue Request Form application, with its library management form, was opened and its dashboard reviewed.
- Two custom permission sets, Library Administrator and Library Member, were added alongside the default Read and Write sets under User Permissions.
- The role hierarchy CEO → {library admin, student, Faculty, Staff} was created.
- A data-sharing rule from student to library admin with Read/Write access on the library management form was configured and saved.

Conclusion

The experiment demonstrated how permissions, roles, role hierarchy and data sharing together form a role-based access control mechanism in Zoho Creator. Structured forms and a role-aware dashboard replace the physical register and verbal requests of a manual library counter with clean, consistent, and analysable data. The two-level hierarchy and the directional Read/Write share rule keep information channels controlled: the library admin gains exactly the access needed to review and process student book issue requests, while the CEO retains overall oversight and the student, Faculty, and Staff roles continue to own their respective requests. The result is a secure, auditable and scalable access-control framework.

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